



8960 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with JWST

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	NGC288a	NIRCam Imaging	(1) NGC-288

Folder	Observation	Label	Observing Template	Science Target
	2	NGC288b	NIRCam Imaging	(1) NGC-288
	3	NGC6723a	NIRCam Imaging	(8) NGC-6723
	4	NGC6723b	NIRCam Imaging	(8) NGC-6723
	5	NGC2808a	NIRCam Imaging	(2) NGC-2808
	6	NGC2808b	NIRCam Imaging	(2) NGC-2808
	13	NGC6553a	NIRCam Imaging	(6) NGC-6553
	14	NGC6553b	NIRCam Imaging	(6) NGC-6553
	7	NGC1851a	NIRCam Imaging	(7) NGC-1851
	8	NGC1851b	NIRCam Imaging	(7) NGC-1851
	9	M22a	NIRCam Imaging	(4) M-22
	10	M22b	NIRCam Imaging	(4) M-22
	11	NGC6528a	NIRCam Imaging	(5) NGC-6528
	12	NGC6528b	NIRCam Imaging	(5) NGC-6528

ABSTRACT

Despite extensive observations, the existence of multiple stellar populations in globular clusters (GCs) remains an enigma in stellar astrophysics. Their formation may conceal exotic mechanisms of star formation occurring in the early Universe and may impact on the cosmic reionization. This proposal advocates for an extensive exploration of multiple populations with the JWST, considering it a step as natural as urgent for piecing together the intricate puzzle of their formation.

Specifically, JWST observations have been successful in detecting multiple stellar populations in the low-mass regime of M dwarfs, difficult to be accessed by previous telescope facilities. The comparison of multiple stellar populations in these stars with those already observed at higher masses, is a crucial ingredient to constrain different scenarios.

We propose a survey of GCs which deeply explores the variegated zoo of multiple stellar populations in the low-mass stars. The comparison of the multiple populations at different masses and in clusters displaying different numbers of stellar populations, possibly experiencing distinct paths in their development and evolution, will represent a crucial piece of knowledge to finally understand the phenomenon, will provide the community with uniform observations of low mass stars in star clusters, and last but not least, promises a unique opportunity to leverage the recent JWST observations of first stellar systems at high redshifts, offering unprecedented insights into the nature of ancient stellar systems.

OBSERVING DESCRIPTION

We propose NIRCам observations in the F115W, F444W and in the F200W F277W filters of seven Galactic globular clusters. The images will be dithered (primary dither type intramodule, and standard subpixel dither type) and will be used to derive stellar photometry and astrometry by means of effective-PSF fitting.

We also propose parallel images with NIRISS in F115W and F200W.

50% of the images will be collected as part as cycle 4 and the remaining 50% of the images of the same field will be collected two years later. These multi-epoch observations will be used to calculate stellar proper motions.

Proposal 8960 - Targets - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with JWST

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	NGC-288	RA: 00 52 27.0000 (13.1125000d) Dec: -26 38 6.10 (-26.63503d) Equinox: J2000	Proper Motion RA: 4.22 mas/yr Proper Motion Dec: -5.65000007100025 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]				
	(2)	NGC-2808	RA: 09 11 39.0000 (137.9125000d) Dec: -64 55 49.00 (-64.93028d) Equinox: J2000	Proper Motion RA: 1.02 mas/yr Proper Motion Dec: 0.28 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]				
	(4)	M-22	RA: 18 36 46.9400 (279.1955833d) Dec: -23 56 31.10 (-23.94197d) Equinox: J2000	Proper Motion RA: 9.82 mas/yr Proper Motion Dec: -5.540000074688578 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]				
	(5)	NGC-6528	RA: 18 04 49.6100 (271.2067083d) Dec: -30 03 20.80 (-30.05578d) Equinox: J2000	Proper Motion RA: -2.157 mas/yr Proper Motion Dec: -5.649000013363548 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]				
	(6)	NGC-6553	RA: 18 09 15.6800 (272.3153333d) Dec: -25 54 27.90 (-25.90775d) Equinox: J2000	Proper Motion RA: 0.3 mas/yr Proper Motion Dec: -0.41000009787239833 mas/yr Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]				
	(7)	NGC-1851	RA: 05 13 44.6000 (78.4358333d) Dec: -40 05 16.00 (-40.08778d) Equinox: J2000	Proper Motion RA: 2.12 mas/yr Proper Motion Dec: -0.6300000904957415 mas/yr Parallax: 8.8E-5" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Stellar Cluster Description=[Globular star clusters]				
	(8)	NGC-6723	RA: 18 59 18.8000 (284.8283333d) Dec: -36 39 15.90 (-36.65442d) Equinox: J2000	Proper Motion RA: 1.0 mas/yr Proper Motion Dec: -2.4199999188567745 mas/yr Parallax: 1.32E-4" Epoch of Position: 2000	
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Stellar Cluster Description=[Globular star clusters]				

Proposal 8960 - Observation 1 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Observation	Proposal 8960, Observation 1: NGC288a										Mon Oct 06 18:00:10 GMT 2025		
	Diagnostic Status: Warning												
	Observing Template: NIRCcam Imaging												
	Coordinated Parallel Template(s): NIRISS Imaging												
Diagnostics	(Visit 1:1) Warning (Form): Data Excess over lower threshold												
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(1)	NGC-288	RA: 00 52 27.0000 (13.1125000d)			Proper Motion RA: 4.22 mas/yr							
			Dec: -26 38 6.10 (-26.63503d)			Proper Motion Dec: -5.65000007100025 mas/yr							
			Equinox: J2000			Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]												
Template	NIRCcam Imaging												
	NIRISS Imaging												
	Module: ALL												
	Subarray: FULL												
	Target Placement: Module Gap												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING	
Spectral Elements	NIRCcam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID			
	1	F115W	F444W	BRIGHT2	6	2	12	6	1610.516				
	2	F200W	F277W	BRIGHT2	5	2	12	6	1352.833				
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	F115W		NIS	6	1	6	6	1610.516				
	2	F200W		NIS	5	1	6	6	1352.833				

Proposal 8960 - Observation 1 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Special Requirements	No Parallel Attachments
	Same Aperture PA 1, 2

Proposal 8960 - Observation 2 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Observation	Proposal 8960, Observation 2: NGC288b										Mon Oct 06 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRCcam Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnostics	(Visit 2:1) Warning (Form): Data Excess over lower threshold											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	NGC-288	RA: 00 52 27.0000 (13.1125000d) Dec: -26 38 6.10 (-26.63503d) Equinox: J2000				Proper Motion RA: 4.22 mas/yr Proper Motion Dec: -5.65000007100025 mas/yr Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]											
Template	NIRCcam Imaging											
	NIRISS Imaging											
	Module: ALL											
	Subarray: FULL Target Placement: Module Gap											
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING
Spectral Elements	NIRCcam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT2	6	2		12	6	1610.516		
	2	F200W	F277W	BRIGHT2	5	2		12	6	1352.833		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F115W		NIS	6	1		6	6	1610.516		
	2	F200W		NIS	5	1		6	6	1352.833		

Special Requirements	No Parallel Attachments Same Aperture PA 1, 2
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Proposal 8960 - Observation 3 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Observation	Proposal 8960, Observation 3: NGC6723a										Mon Oct 06 18:00:10 GMT 2025		
	Diagnostic Status: Warning												
	Observing Template: NIRCам Imaging												
	Coordinated Parallel Template(s): NIRISS Imaging												
Diagnostics	(Visit 3:1) Warning (Form): Data Excess over lower threshold												
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous				
	(8)	NGC-6723	RA: 18 59 18.8000 (284.8283333d) Dec: -36 39 15.90 (-36.65442d) Equinox: J2000				Proper Motion RA: 1.0 mas/yr Proper Motion Dec: -2.4199999188567745 mas/yr Parallax: 1.32E-4" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.												
	Category=Stellar Cluster Description=[Globular star clusters]												
Template	NIRCам Imaging												
	NIRISS Imaging												
	Module: ALL												
	Subarray: FULL Target Placement: Module Gap												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING	
Spectral Elements	NIRCам Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F115W	F444W	BRIGHT2	6	2		12	6	1610.516			
	2	F200W	F277W	BRIGHT2	5	2		12	6	1352.833			
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	F115W		NIS	6	1		6	6	1610.516			
	2	F200W		NIS	5	1		6	6	1352.833			

Special Requirements	No Parallel Attachments Same Aperture PA 3, 4
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Proposal 8960 - Observation 4 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Observation	Proposal 8960, Observation 4: NGC6723b										Mon Oct 06 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRCам Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnostics	(Visit 4:1) Warning (Form): Data Excess over lower threshold											
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(8)	NGC-6723	RA: 18 59 18.8000 (284.8283333d) Dec: -36 39 15.90 (-36.65442d) Equinox: J2000			Proper Motion RA: 1.0 mas/yr Proper Motion Dec: -2.4199999188567745 mas/yr Parallax: 1.32E-4" Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Stellar Cluster Description=[Globular star clusters]											
Template	NIRCам Imaging											
	NIRISS Imaging											
	Module: ALL											
	Subarray: FULL											
Dithers	Target Placement: Module Gap											
	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING
Spectral Elements	NIRCам Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT2	6	2		12	6	1610.516		
	2	F200W	F277W	BRIGHT2	5	2		12	6	1352.833		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F115W		NIS	6	1		6	6	1610.516		
	2	F200W		NIS	5	1		6	6	1352.833		

Special Requirements	No Parallel Attachments Same Aperture PA 3, 4
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Proposal 8960 - Observation 5 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Observation	Proposal 8960, Observation 5: NGC2808a										Mon Oct 06 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRCIm Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnostics	(Visit 5:1) Warning (Form): Data Excess over lower threshold											
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous			
	(2)	NGC-2808	RA: 09 11 39.0000 (137.9125000d) Dec: -64 55 49.00 (-64.93028d) Equinox: J2000				Proper Motion RA: 1.02 mas/yr Proper Motion Dec: 0.28 mas/yr Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	Category=Stellar Cluster Description=[Globular star clusters]											
Template	NIRCIm Imaging											
	NIRISS Imaging											
	Module: ALL											
	Subarray: FULL Target Placement: Module Gap											
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING
Spectral Elements	NIRCIm Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT2	8	2		12	6	2125.88		
	2	F200W	F277W	BRIGHT2	6	2		12	6	1610.516		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F115W		NIS	8	1		6	6	2125.88		
	2	F200W		NIS	6	1		6	6	1610.516		

Proposal 8960 - Observation 5 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Special Requirements	Aperture PA Range 229.92542306 to 139.92542306 Degrees (V3 230.0 to 140.0) No Parallel Attachments Same Aperture PA 5, 6
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Proposal 8960 - Observation 6 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Observation	Proposal 8960, Observation 6: NGC2808b										Mon Oct 06 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRCcam Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnostics	(Visit 6:1) Warning (Form): Data Excess over lower threshold											
	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	NGC-2808	RA: 09 11 39.0000 (137.9125000d)			Proper Motion RA: 1.02 mas/yr						
			Dec: -64 55 49.00 (-64.93028d)			Proper Motion Dec: 0.28 mas/yr						
			Equinox: J2000			Epoch of Position: 2000						
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]									
Template	NIRCcam Imaging											
	NIRISS Imaging											
	Module: ALL											
	Subarray: FULL											
	Target Placement: Module Gap											
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING
Spectral Elements		NIRCcam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1		F115W	F444W	BRIGHT2	8	2	12	6	2125.88		
	2		F200W	F277W	BRIGHT2	6	2	12	6	1610.516		
Spectral Elements		NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		F115W		NIS	8	1	6	6	2125.88		
	2		F200W		NIS	6	1	6	6	1610.516		

Special Requirements	No Parallel Attachments Same Aperture PA 5, 6
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Proposal 8960 - Observation 13 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations wit...

Observation	Proposal 8960, Observation 13: NGC6553a										Mon Oct 06 18:00:10 GMT 2025		
	Diagnostic Status: Warning												
	Observing Template: NIRCcam Imaging												
	Coordinated Parallel Template(s): NIRISS Imaging												
Diagnostics	(Visit 13:1) Warning (Form): Data Excess over lower threshold												
	(Visit 13:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous				
	(6)	NGC-6553	RA: 18 09 15.6800 (272.3153333d)			Proper Motion RA: 0.3 mas/yr							
			Dec: -25 54 27.90 (-25.90775d)			Proper Motion Dec: -0.41000009787239833 mas/yr							
			Equinox: J2000			Epoch of Position: 2000							
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Stellar Cluster												
	Description=[Globular star clusters]												
Template	NIRCcam Imaging					NIRISS Imaging							
	Module: ALL												
	Subarray: FULL												
	Target Placement: Module Gap												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING	
Spectral Elements	NIRCcam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F115W	F444W	BRIGHT2	8	2		12	6	2125.88			
	2	F200W	F277W	BRIGHT2	6	2		12	6	1610.516			
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	F115W		NIS	8	1		6	6	2125.88			
	2	F200W		NIS	6	1		6	6	1610.516			

Special Requirements	No Parallel Attachments Same Aperture PA 13, 14
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Proposal 8960 - Observation 14 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations wit...

Observation	Proposal 8960, Observation 14: NGC6553b										Mon Oct 06 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRCcam Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnostics	(Visit 14:1) Warning (Form): Data Excess over lower threshold											
	(Visit 14:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(6)	NGC-6553	RA: 18 09 15.6800 (272.3153333d) Dec: -25 54 27.90 (-25.90775d) Equinox: J2000			Proper Motion RA: 0.3 mas/yr Proper Motion Dec: -0.41000009787239833 mas/yr Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	Category=Stellar Cluster Description=[Globular star clusters]											
Template	NIRCcam Imaging					NIRISS Imaging						
	Module: ALL											
	Subarray: FULL											
	Target Placement: Module Gap											
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING
Spectral Elements	NIRCcam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT2	8	2		12	6	2125.88		
	2	F200W	F277W	BRIGHT2	6	2		12	6	1610.516		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F115W		NIS	8	1		6	6	2125.88		
	2	F200W		NIS	6	1		6	6	1610.516		

Special Requirements	No Parallel Attachments Same Aperture PA 13, 14
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Proposal 8960 - Observation 7 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Observation	Proposal 8960, Observation 7: NGC1851a										Mon Oct 06 18:00:10 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRCам Imaging										
	Coordinated Parallel Template(s): NIRISS Imaging										
Diagnostics	(Visit 7:1) Warning (Form): Data Excess over lower threshold										
	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	NGC-1851	RA: 05 13 44.6000 (78.4358333d) Dec: -40 05 16.00 (-40.08778d) Equinox: J2000			Proper Motion RA: 2.12 mas/yr Proper Motion Dec: -0.6300000904957415 mas/yr Parallax: 8.8E-5" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.										
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.										
	Category=Stellar Cluster Description=[Globular star clusters]										
Template	NIRCам Imaging										
	NIRISS Imaging										
	Module: ALL										
	Subarray: FULL										
Target Placement: Module Gap											
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector	Dither Direct Images Primes
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS	NO_DITHERING
Spectral Elements	NIRCам Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F115W	F444W	BRIGHT2	7	2		12	6	1868.198	
	2	F200W	F277W	BRIGHT2	5	2		12	6	1352.833	
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	F115W		NIS	7	1		6	6	1868.198	
	2	F200W		NIS	5	1		6	6	1352.833	

Special Requirements	No Parallel Attachments Same Aperture PA 7, 8
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Proposal 8960 - Observation 8 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Observation	Proposal 8960, Observation 8: NGC1851b										Mon Oct 06 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRCam Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnostics	(Visit 8:1) Warning (Form): Data Excess over lower threshold											
	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous			
	(7)	NGC-1851	RA: 05 13 44.6000 (78.4358333d) Dec: -40 05 16.00 (-40.08778d) Equinox: J2000				Proper Motion RA: 2.12 mas/yr Proper Motion Dec: -0.6300000904957415 mas/yr Parallax: 8.8E-5" Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.											
	Category=Stellar Cluster Description=[Globular star clusters]											
Template	NIRCam Imaging											
	NIRISS Imaging											
	Module: ALL											
	Subarray: FULL											
Target Placement: Module Gap												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT2	7	2		12	6	1868.198		
	2	F200W	F277W	BRIGHT2	5	2		12	6	1352.833		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F115W		NIS	7	1		6	6	1868.198		
	2	F200W		NIS	5	1		6	6	1352.833		

Special Requirements	No Parallel Attachments Same Aperture PA 7, 8
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Proposal 8960 - Observation 9 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Observation	Proposal 8960, Observation 9: M22a										Mon Oct 06 18:00:10 GMT 2025		
	Diagnostic Status: Warning												
	Observing Template: NIRCcam Imaging												
	Coordinated Parallel Template(s): NIRISS Imaging												
Diagnostics	(Visit 9:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(4)	M-22	RA: 18 36 46.9400 (279.1955833d)				Proper Motion RA: 9.82 mas/yr			Proper Motion Dec: -5.540000074688578 mas/yr Epoch of Position: 2000			
	Dec: -23 56 31.10 (-23.94197d)												
	Equinox: J2000												
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]													
Template	NIRCcam Imaging											NIRISS Imaging	
	Module: ALL												
	Subarray: FULL												
	Target Placement: Module Gap												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING	
Spectral Elements	NIRCcam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID			
	1	F115W	F444W	BRIGHT2	6	1	6	6	773.047				
	2	F200W	F277W	BRIGHT2	5	1	6	6	644.206				
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID			
	1	F115W		NIS	3	1	6	6	837.468				
	2	F200W		NIS	2	1	6	6	579.786				

Proposal 8960 - Observation 9 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations with ...

Special Requirements	Aperture PA Range 80 to 95 Degrees (V3 80.07457694 to 95.07457694) No Parallel Attachments Same Aperture PA 9, 10
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Proposal 8960 - Observation 10 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations wit...

Observation	Proposal 8960, Observation 10: M22b										Mon Oct 06 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRCcam Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnos	(Visit 10:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name		Target Coordinates			Targ. Coord. Corrections		Miscellaneous			
	(4)	M-22		RA: 18 36 46.9400 (279.1955833d)			Proper Motion RA: 9.82 mas/yr					
				Dec: -23 56 31.10 (-23.94197d)			Proper Motion Dec: -5.540000074688578 mas/yr					
				Equinox: J2000			Epoch of Position: 2000					
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Stellar Cluster Description=[Globular star clusters]											
Template	NIRCcam Imaging					NIRISS Imaging						
	Module: ALL											
	Subarray: FULL											
	Target Placement: Module Gap											
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING
Spectral Elements	NIRCcam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT2	6	1		6	6	773.047		
	2	F200W	F277W	BRIGHT2	5	1		6	6	644.206		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F115W		NIS	3	1		6	6	837.468		
	2	F200W		NIS	2	1		6	6	579.786		

Special Requirements	No Parallel Attachments Same Aperture PA 9, 10
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Proposal 8960 - Observation 11 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations wit...

Observation	Proposal 8960, Observation 11: NGC6528a										Mon Oct 06 18:00:10 GMT 2025		
	Diagnostic Status: Warning												
	Observing Template: NIRCam Imaging												
	Coordinated Parallel Template(s): NIRISS Imaging												
Diagnostics	(Visit 11:1) Warning (Form): Data Excess over lower threshold												
	(Visit 11:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(5)	NGC-6528	RA: 18 04 49.6100 (271.2067083d) Dec: -30 03 20.80 (-30.05578d) Equinox: J2000				Proper Motion RA: -2.157 mas/yr Proper Motion Dec: -5.649000013363548 mas/yr Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Stellar Cluster Description=[Globular star clusters]												
Template	NIRCam Imaging												
	NIRISS Imaging												
	Module: ALL												
	Subarray: FULL Target Placement: Module Gap												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING	
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F115W	F444W	BRIGHT2	9	2		12	6	2383.563			
	2	F200W	F277W	BRIGHT2	7	2		12	6	1868.198			
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	F115W		NIS	9	1		6	6	2383.563			
	2	F200W		NIS	7	1		6	6	1868.198			

Special Requirements	No Parallel Attachments Same Aperture PA 11, 12
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Proposal 8960 - Observation 12 - A survey of low mass stars in globular clusters: Probing the Origin of Multiple Stellar Populations wit...

Observation	Proposal 8960, Observation 12: NGC6528b										Mon Oct 06 18:00:10 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: NIRCам Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnostics	(Visit 12:1) Warning (Form): Data Excess over lower threshold											
	(Visit 12:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(5)	NGC-6528	RA: 18 04 49.6100 (271.2067083d) Dec: -30 03 20.80 (-30.05578d) Equinox: J2000			Proper Motion RA: -2.157 mas/yr Proper Motion Dec: -5.649000013363548 mas/yr Epoch of Position: 2000						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	Category=Stellar Cluster Description=[Globular star clusters]											
Template	NIRCam Imaging											
	NIRISS Imaging											
	Module: ALL											
	Subarray: FULL											
Target Placement: Module Gap												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	INTRAMODULEBOX		3				1		2-POINT-LARGE-WITH-NIRISS		NO_DITHERING
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F115W	F444W	BRIGHT2	9	2		12	6	2383.563		
	2	F200W	F277W	BRIGHT2	7	2		12	6	1868.198		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F115W		NIS	9	1		6	6	2383.563		
	2	F200W		NIS	7	1		6	6	1868.198		

Special Requirements	No Parallel Attachments Same Aperture PA 11, 12
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